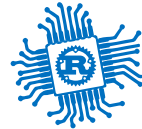


# Exceptions and Interrupts

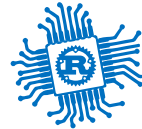
Lecture 9



# Exceptions and Interrupts

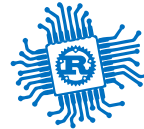
used by RP2

- Exceptions
- Interrupts
- Boot



# Exceptions

for the ARM Cortex-M processor

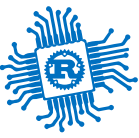


# Bibliography

for this section

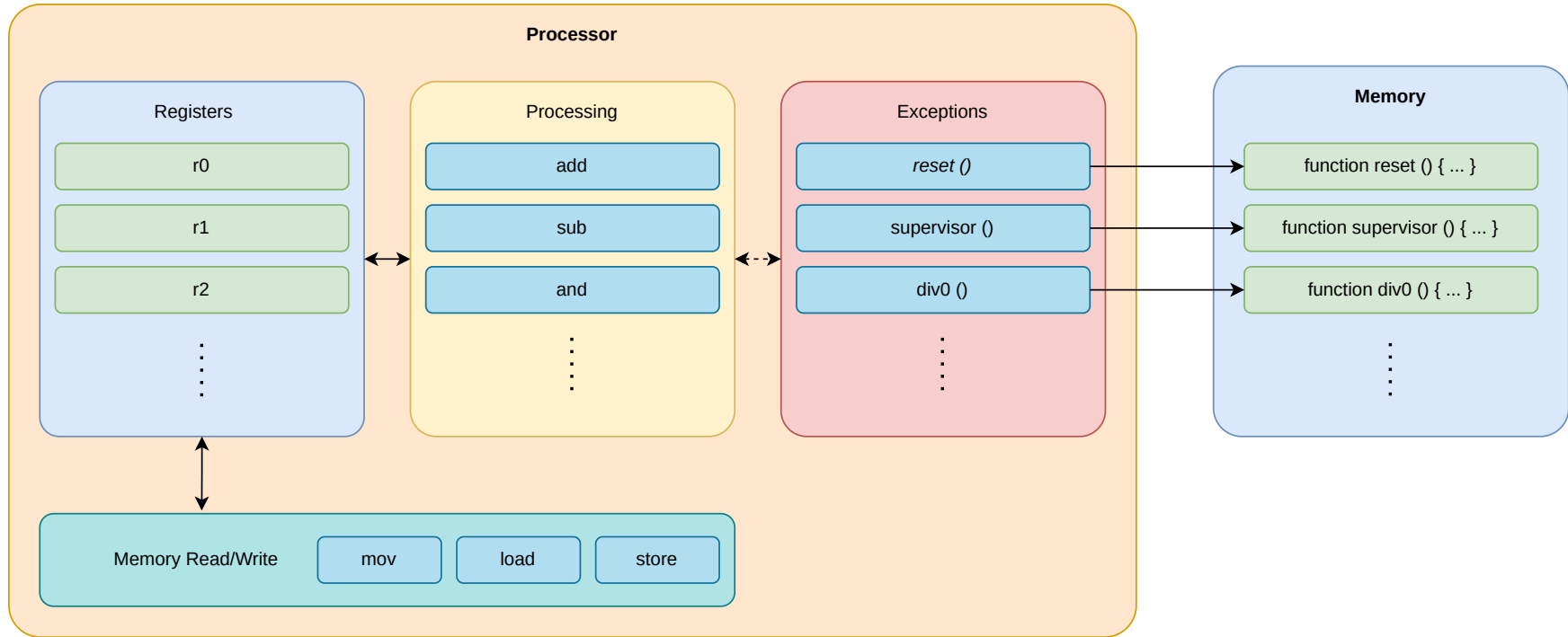
**Joseph Yiu**, *The Definitive Guide to ARM® Cortex®-M23 and Cortex-M33 Processors*

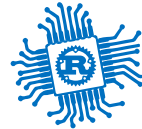
- Chapter 4 - *Architecture*
  - Section 4.5 - *Exceptions and Interrupts*
    - Subsection 4.4.1 - *What are exceptions*
- Chapter 8 - *Exceptions and Interrupts*
  - Section 8.1 - *What are Exceptions and Interrupts*
  - Section 8.2 - *Exception types+*



# Processor Exceptions

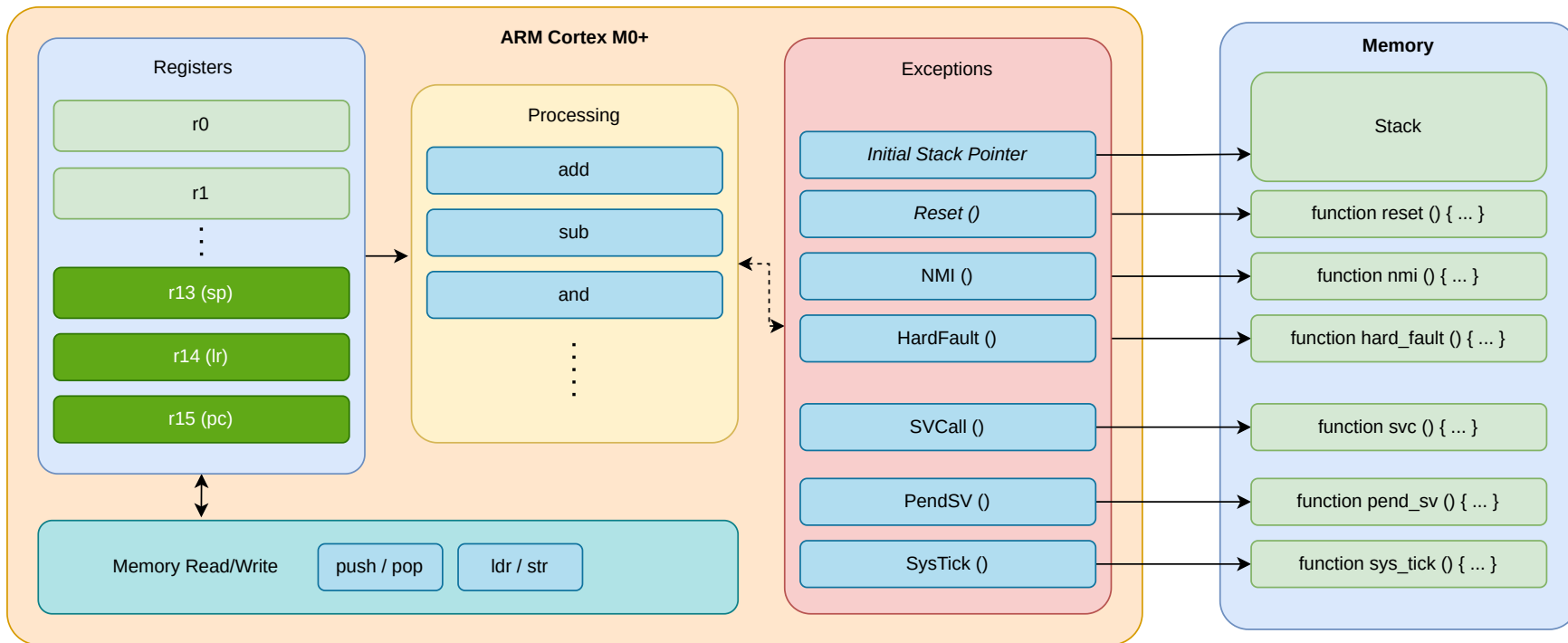
what happens if something does not work as required

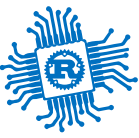




# ARM Cortex-M Exceptions

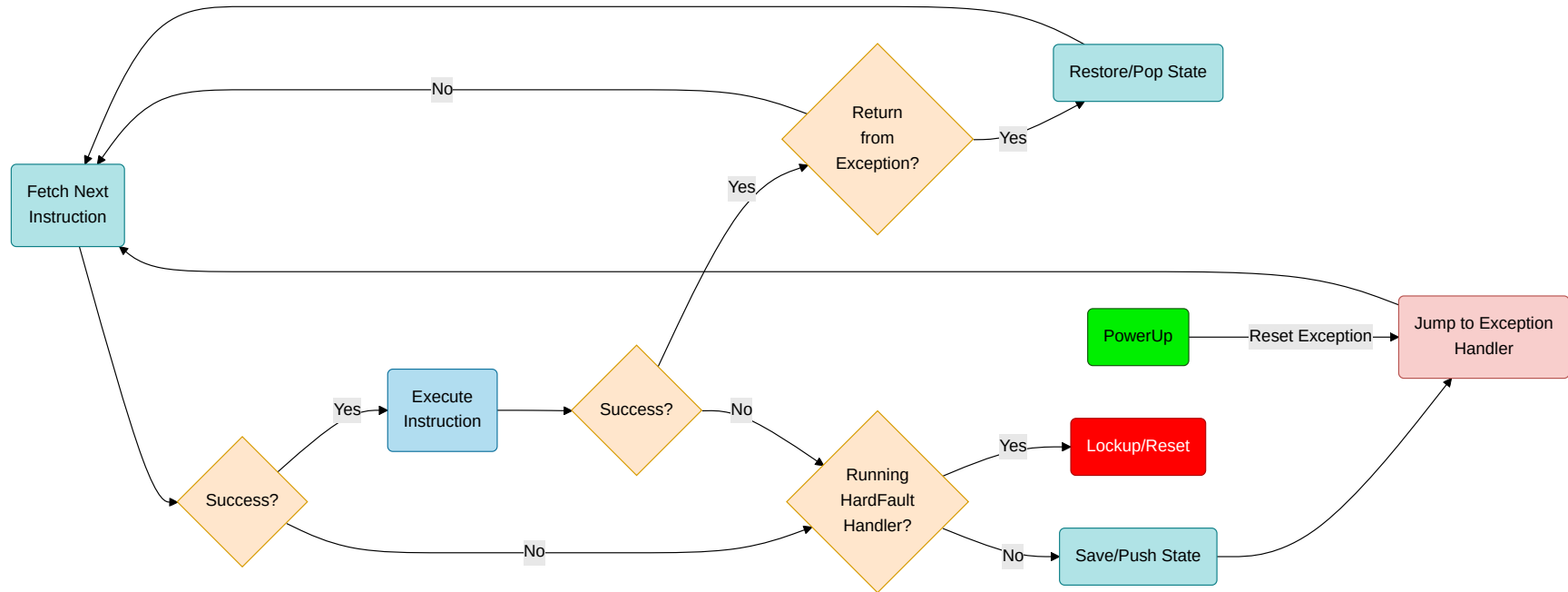
what happens if something does not work as required



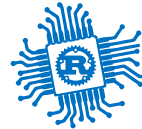


# Exception (HardFault) Handling

ARM Cortex-M has one **actual** exception, *HardFault*

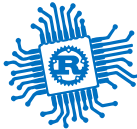


- the exception table of RP2040 at address 0x1000\_0100, RP2350 at address 0x1000\_0000
- the processor generates a *Reset* exception when it starts



# Interrupts

for ARM Cortex-M0+



# Bibliography

for this section

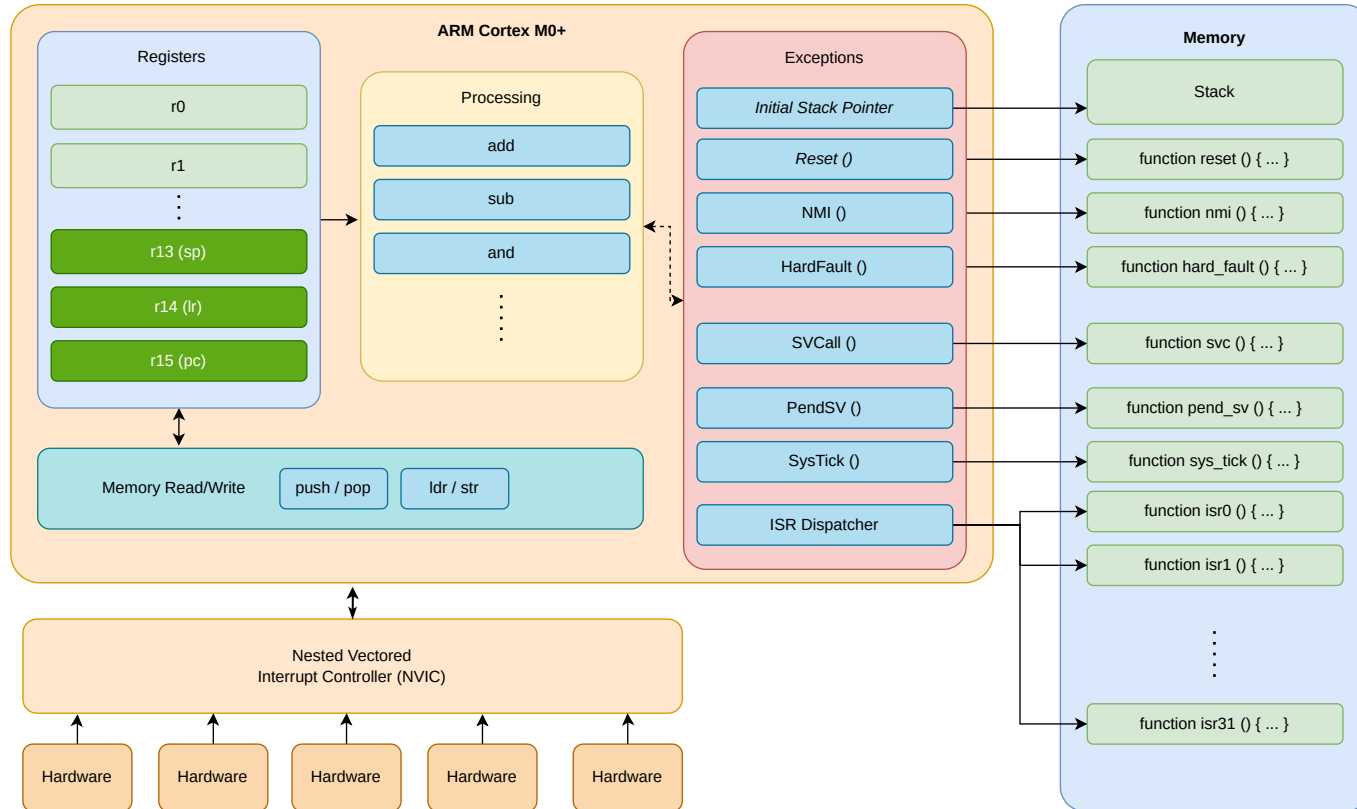
**Joseph Yiu**, *The Definitive Guide to ARM® Cortex®-M0 and Cortex-M0+ Processors, 2nd Edition*

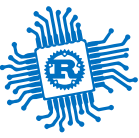
- Chapter 8 - *Exceptions and Interrupts*
  - Section 8.1 - *What are Exceptions and Interrupts*
  - Section 8.3 - *Brief Overview of the NVIC*
  - Section 8.4 - *Definition of Exception Priority Levels*
  - Section 8.5 - *Vector Table*
  - Section 8.6 - *Exception Sequence Overview*
- Chapter 11 - *Fault Handling*
  - Section 11.1 - *Fault Exception Overview*
  - Section 11.2 - *What Can Cause a Fault*
  - Section 11.7 - *Lockup*



# ARM Cortex-M Interrupts

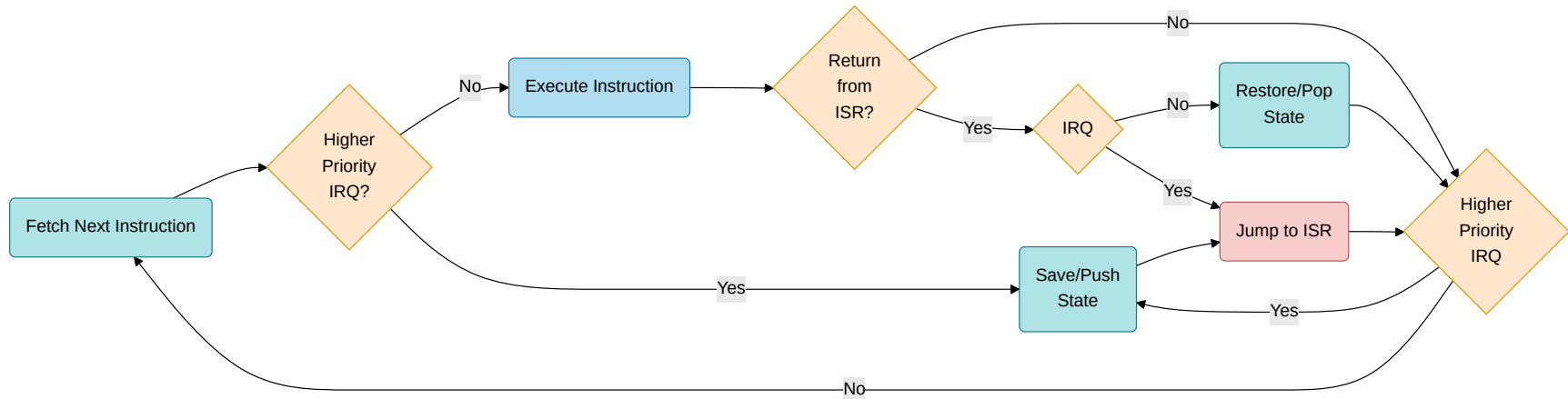
some hardware device notifies the MCU





# Interrupt Handling

ARM Cortex-M



*IRQ*      Interrupt Request

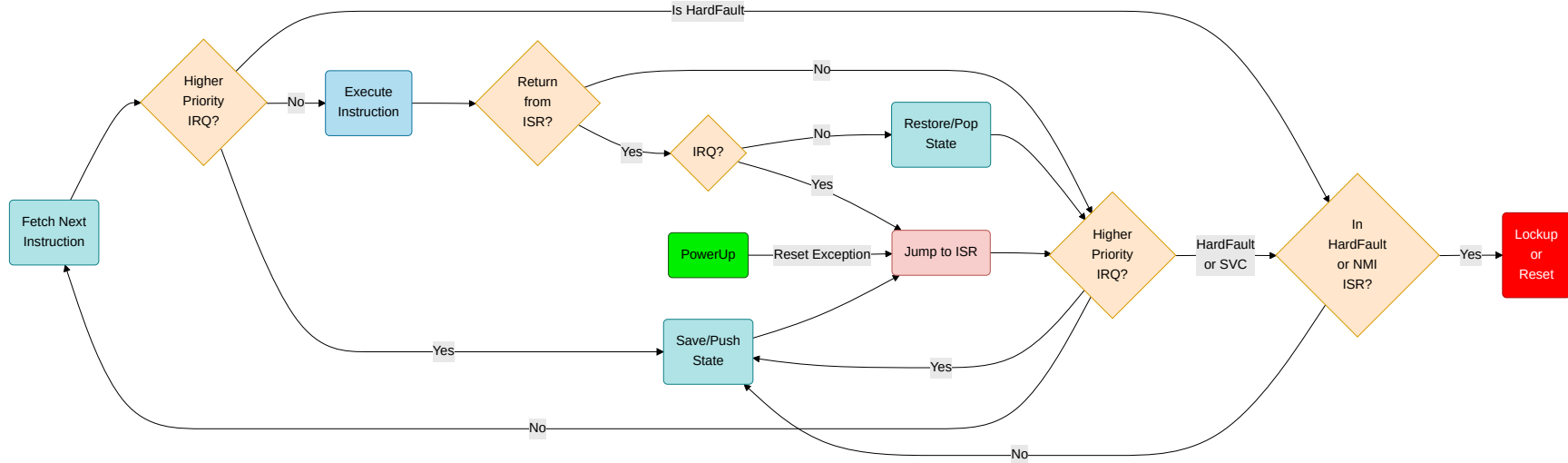
*ISR*      Interrupt Service Routine

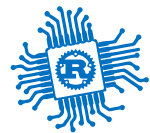
- the interrupt vector (table) of RP2350 starts at address 0x1000\_0040 (after the exceptions table with 15 interrupts)
- ARM Cortex-M0+ has a maximum of 32 IRQs
- ARM Cortex-M33 has a maximum of 480 IRQs



# Exceptions are Software Interrupt Requests

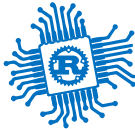
with a negative IRQ number and a higher priority





# Boot

of the RP2040 and RP2350



# Bibliography

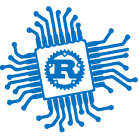
for this section

## **Raspberry Pi Ltd, RP2040 Datasheet**

- Chapter 2 - *System Description*
  - Section 2.7 - *Boot sequence*
  - Section 2.8 - *Bootrom*
    - Subsection 2.8.1 - *Processor Controlled Boot Sequence*

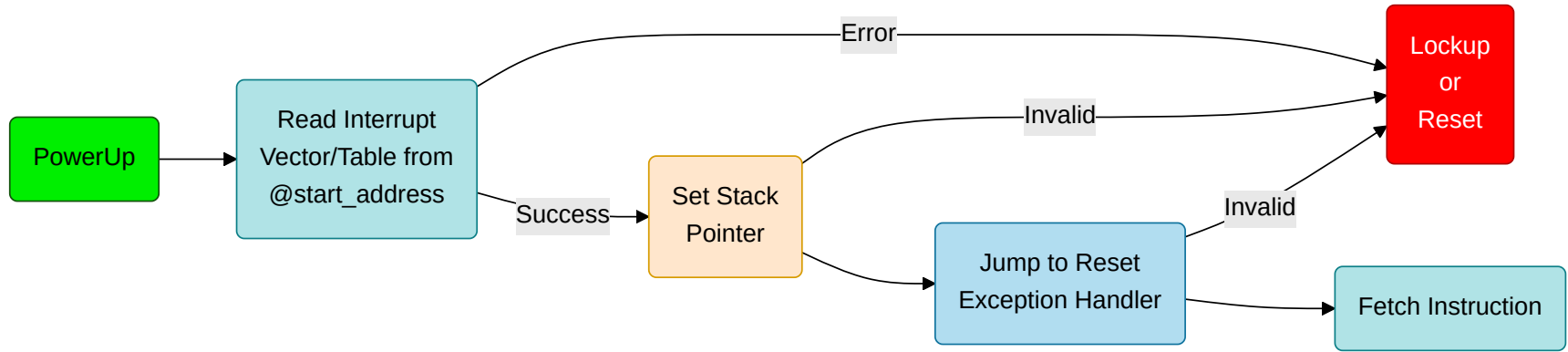
## **Raspberry Pi Ltd, RP2350 Datasheet**

- Chapter 5 - *Bootrom*
  - Section 5.1 - *Bootrom concepts*



# Boot

how the ARM Cortex-M starts

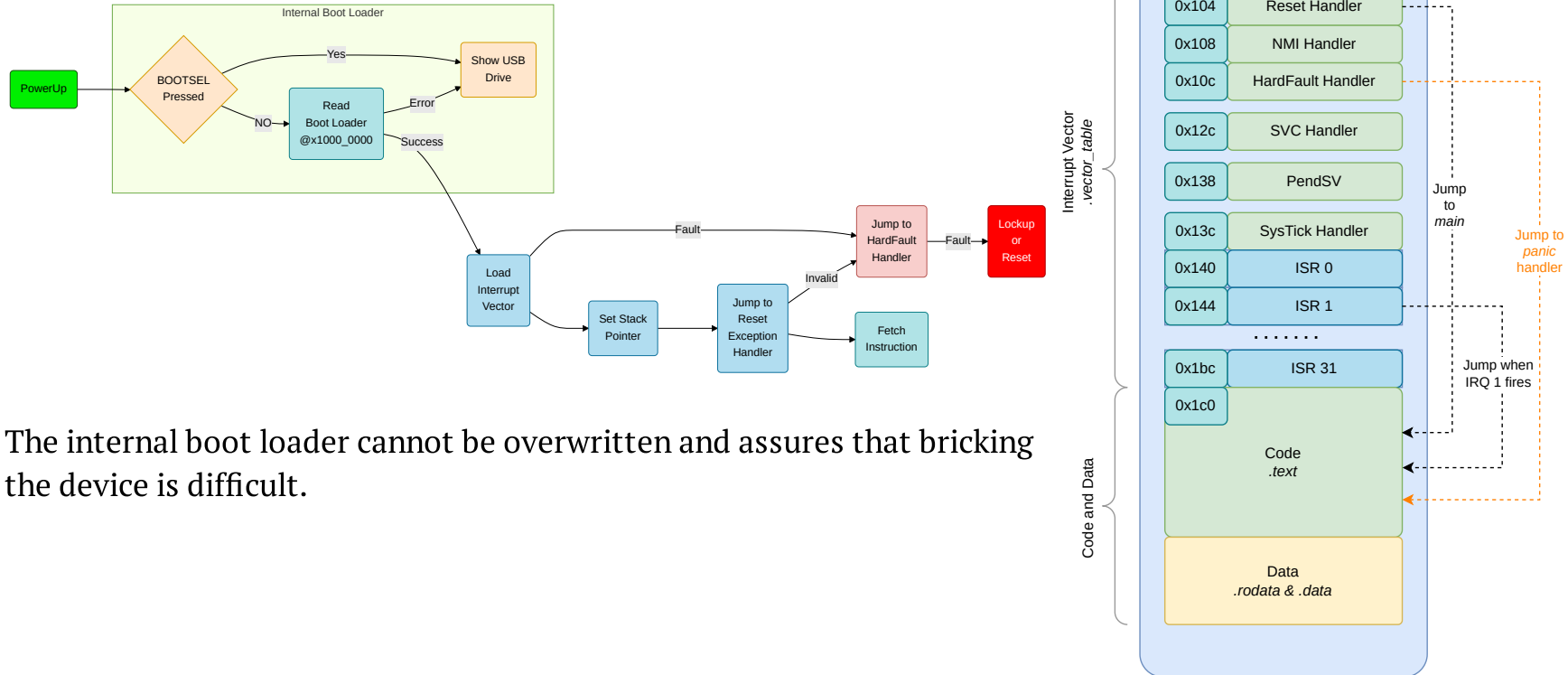


- the *start\_address* for RP2040 is 0x1000\_0100
- the *start\_address* for RP2350 depends on the *Start Block* items
- RP2040 & RP2350 have another boot loader that it loads from 0x0000\_0000



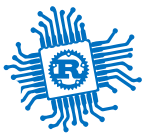
# Boot

## The RP2040 boot process



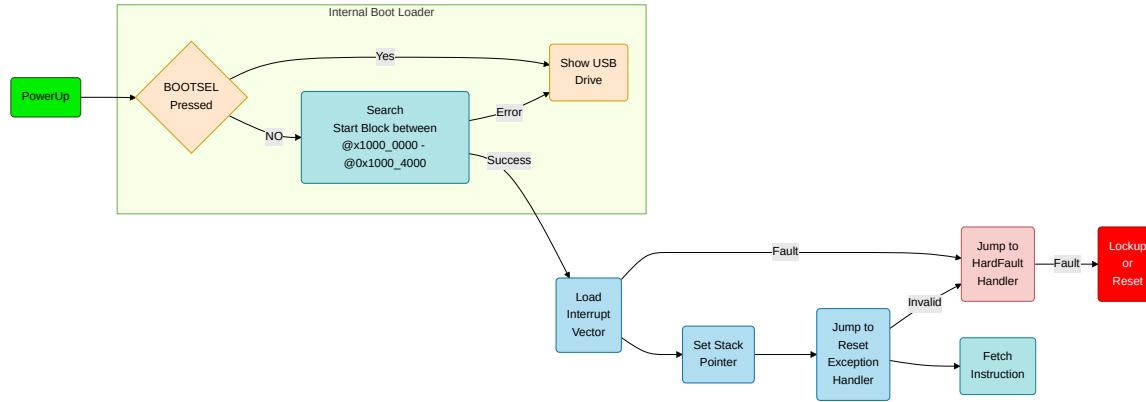
\* drawing is not at scale, code and data are significantly greater than the interrupt vector

The internal boot loader cannot be overwritten and assures that bricking the device is difficult.

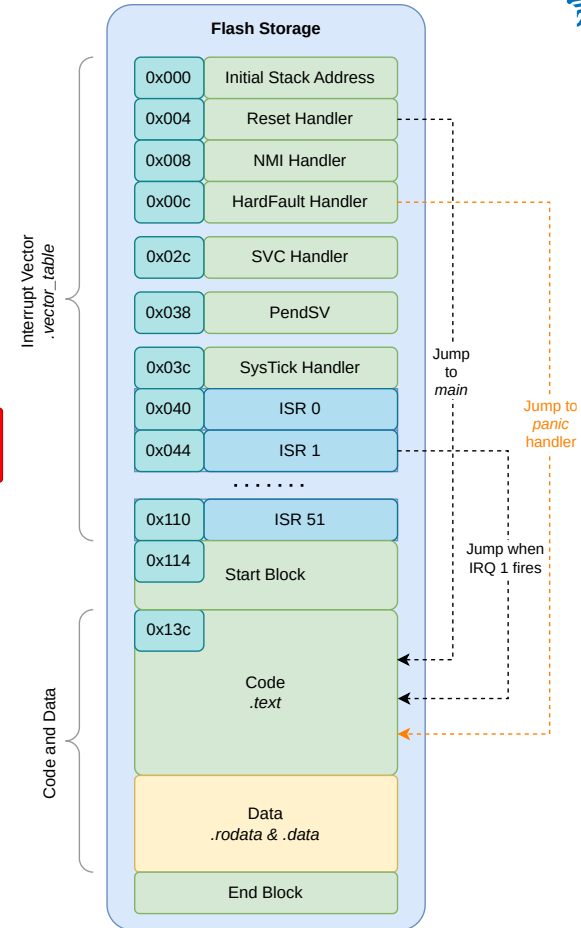


# Boot

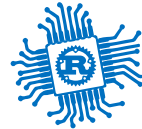
## The RP2350 boot process



The internal boot loader cannot be overwritten and assures that bricking the device is difficult.



\* drawing is not at scale, code and data are significantly greater than the interrupt vector



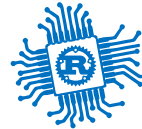
# Set Fault Handler

bare metal, pac or embassy-rs

```
// defined by the cortex-m-rt crate
pub struct ExceptionFrame {
    r0: u32,
    r1: u32,
    r2: u32,
    r3: u32,
    r12: u32,
    lr: u32,
    pc: u32,
    xpsr: u32,
}
```

**HardFault** never returns

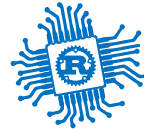
```
1  #[exception]
2  unsafe fn HardFault(_frame: &ExceptionFrame) -> ! {
3      panic!("HardFault {:?}", frame);
4  }
```



# Set SysTick Handler

bare metal, PAC or embassy-rs

```
1  #[exception]
2  unsafe fn SysTick() {
3      // execute at a fixed interval
4  }
```

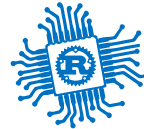


# Set Interrupt Handlers

*embassy-rs already defined the interrupts as it needs them*

| IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source |
|-----|------------------|-----|------------------|-----|------------------|-----|------------------|-----|------------------|
| 0   | TIMER_IRQ_0      | 6   | XIP_IRQ          | 12  | DMA_IRQ_1        | 18  | SPI0_IRQ         | 24  | I2C1_IRQ         |
| 1   | TIMER_IRQ_1      | 7   | PIO0_IRQ_0       | 13  | IO_IRQ_BANK0     | 19  | SPI1_IRQ         | 25  | RTC_IRQ          |
| 2   | TIMER_IRQ_2      | 8   | PIO0_IRQ_1       | 14  | IO_IRQ_QSPI      | 20  | UART0_IRQ        |     |                  |
| 3   | TIMER_IRQ_3      | 9   | PIO1_IRQ_0       | 15  | SIO_IRQ_PROC0    | 21  | UART1_IRQ        |     |                  |
| 4   | PWM_IRQ_WRAP     | 10  | PIO1_IRQ_1       | 16  | SIO_IRQ_PROC1    | 22  | ADC_IRQ_FIFO     |     |                  |
| 5   | USBCTRL_IRQ      | 11  | DMA_IRQ_0        | 17  | CLOCKS_IRQ       | 23  | I2C0_IRQ         |     |                  |

```
1 // The name of the function has to match the
2 // interrupt name defined by the
3 // *Peripheral Access Crate* (PAC)
4 unsafe fn IO_IRQ_BANK0() {
5     // so some work when a pin interrupt triggers
6 }
```

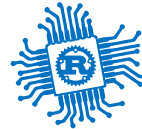


# Use interrupts in embassy-rs

embassy-rs registers interrupt handlers and exposes a high level API

| IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source | IRQ | Interrupt Source |
|-----|------------------|-----|------------------|-----|------------------|-----|------------------|-----|------------------|
| 0   | TIMER_IRQ_0      | 6   | XIP_IRQ          | 12  | DMA_IRQ_1        | 18  | SPI0_IRQ         | 24  | I2C1_IRQ         |
| 1   | TIMER_IRQ_1      | 7   | PI00_IRQ_0       | 13  | IO_IRQ_BANK0     | 19  | SPI1_IRQ         | 25  | RTC_IRQ          |
| 2   | TIMER_IRQ_2      | 8   | PI00_IRQ_1       | 14  | IO_IRQ_QSPI      | 20  | UART0_IRQ        |     |                  |
| 3   | TIMER_IRQ_3      | 9   | PI01_IRQ_0       | 15  | SIO_IRQ_PROC0    | 21  | UART1_IRQ        |     |                  |
| 4   | PWM_IRQ_WRAP     | 10  | PI01_IRQ_1       | 16  | SIO_IRQ_PROC1    | 22  | ADC_IRQ_FIFO     |     |                  |
| 5   | USBCTRL_IRQ      | 11  | DMA_IRQ_0        | 17  | CLOCKS_IRQ       | 23  | I2C0_IRQ         |     |                  |

```
1  #[embassy_executor::main]
2  async fn main(_spawner: Spawner) {
3      let p = embassy_rp::init(Default::default());
4      let mut button = Input::new(p.PIN_20, Pull::None);
5
6      loop {
7          info!("Waiting for the button press");
8
9          // waits for interrupt (sent by button)
10         // IO_IRQ_BANK0
11         button.wait_for_high().await;
12
13         info!("Button was pressed");
14     }
15 }
```



# Conclusion

we talked about

- Exceptions
- Interrupts
- How the RP2040 and RP2350 boot and load the software